

HAT-P-12b

R-0858

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250625 · created 2026-07-28T21:37:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460851.7465 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.052
Transit depth (Rp/Rs)^2	3.1 +/- 1.84 [%]
Orbital Inclination (inc)	85.84 +/- 1.67
Airmass coefficient 1 (a1)	0.951 +/- 0.052
Airmass coefficient 2 (a2)	0.042 +/- 0.042
Scatter in the residuals of the lightcurve fit is	5.43 %
Best Comparison Star	#1 - [367, 182]
Optimal Aperture	4.69
Optimal Annulus	19.36
Transit Duration (day)	0.061 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.052 vs 0.1406 (deviation 0.68σ)
Duration fit vs published	0.061 d vs 0.0975 d (deviation 1.07σ)
Mid-time O–C	12.2 min
Depth SNR	3.4
Residual scatter	5.43 %

Light curve

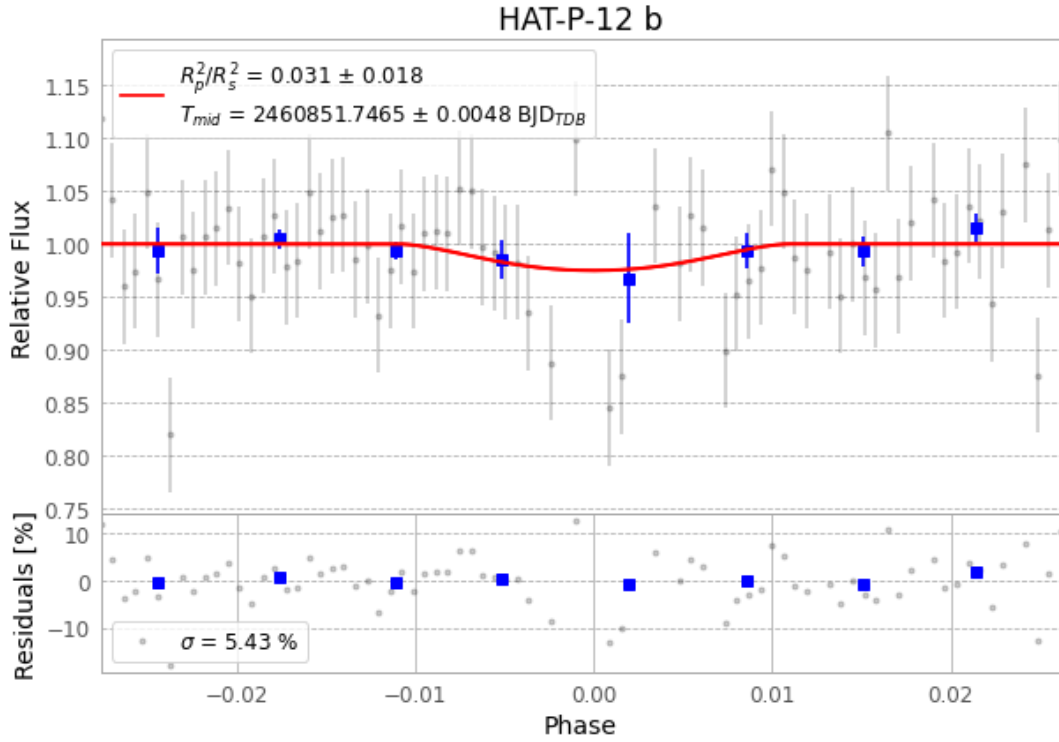


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-06-24.png (SHA-256 c0ab1b0cf24daa12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60e7a9fb2f234d18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), HATP-12250625034516.FITS → HATP-12250625075415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250625.log (de1d94010eef...), FinalLightCurve_HAT-P-12 b_2025-06-24.png (c0ab1b0cf24d...), FinalLightCurve_HAT-P-12 b_2025-06-24.csv (8ee427763934...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:37Z.